MATH1073: Calculus I Quiz# 1 Solution

1. (i) lub=3, glb=-1

(ii) lub=$\frac{\sqrt{5}-1}{2}$, glb=$\frac{-\sqrt{5}-1}{2}$

(iii) lub=1, glb=0

(iv) lub=2, glb=0

(v) lub=$\frac{3}{2}$, glb=-1

2. Given any $\epsilon > 0$, choose $N = \lceil \frac{1}{\epsilon} \rceil$. Then, for all $n \geq N$, we have $\left| \frac{n}{n+1} - 1 \right| = \frac{1}{n+1} < \frac{1}{N} \leq \epsilon$. Hence, $\lim_{n\to\infty} a_n = 1$.

3. (a) $a_n = \cos\left(\frac{n\pi}{n+1}\right) = \cos\left(\frac{n\pi/n}{(n+1)/n}\right) = \cos\left(\frac{\pi}{1+1/n}\right)$ so $a_n \to \cos\pi = -1$ as $n \to \infty$ since $\lim_{n\to\infty} 1/n = 0$. Converges.

(b) $a_n = \frac{n^2}{\sqrt{n^3+4n}} = \frac{n^2/\sqrt{n^3}}{\sqrt{n^3+4n}/\sqrt{n^3}} = \frac{\sqrt{n}}{\sqrt{1+4/n^2}}$, so $a_n \to \infty$ as $n \to \infty$ since $\lim_{n\to\infty} \sqrt{n} = \infty$ and $\lim_{n\to\infty} \sqrt{1+4/n^2} = 1$. Diverges

(c) $a_n = \frac{(2n-1)!}{(2n+1)!} = \frac{(2n-1)!}{(2n+1)(2n)(2n-1)!} = \frac{1}{(2n+1)(2n)} \to 0$ as $n \to \infty$. Converges

(d) $a_n = \frac{\ln n}{\ln 2n} = \frac{\ln n}{\ln 2 + \ln n} = \frac{1}{\frac{\ln 2}{\ln n}+1} \to \frac{1}{0+1} = 1$ as $n \to \infty$. Converges

(e) $0 < |a_n| = \frac{3^n}{n!} = \frac{3}{1}\cdot\frac{3}{2}\cdot\frac{3}{3}\cdot\cdots\cdot\frac{3}{(n-1)}\cdot\frac{3}{n} \leq \frac{3}{1}\cdot\frac{3}{2}\cdot\frac{3}{n}$ [for $n > 3$] $= \frac{27}{2n} \to 0$ as $n \to \infty$, so by the Squeeze Theorem,$\{(-3)^n/n!\}$ converges to 0 .

(f) $0 < a_n = \frac{1}{n}\left(\frac{2\cdot 3\cdot\cdots\cdot n}{n\cdot n\cdot\cdots\cdot n}\right) \leq \frac{1}{n}$, we know that $1/n \to 0$ as $n \to \infty$. Therefore $a_n \to 0$ as $n \to \infty$ by the Squeeze Theorem.

4. We use mathematical induction to show that $a_{n+1} > a_n$ for all $n \geq 1$. This is true for $n = 1$ because $a_2 = 4 > a_1$. If we assume that it is true for $n = k$, then we have so

$$a_{k+1} > a_k$$

and

$$\frac{1}{2}(a_{k+1} + 6) > \frac{1}{2}(a_k + 6)$$

Thus

$$a_{k+2} > a_{k+1}$$

We have deduced that $a_{n+1} > a_n$ is true for $n = k + 1$. Therefore the inequality is true for all n by induction.

Next we verify that $\{a_n\}$ is bounded by showing that $a_n < 6$ for all n. (Since the sequence is increasing, we already know that it has a lower bound: $a_n \geq a_1 = 2$ for all n.) We know that $a_1 < 6$, so the assertion is true for $n = 1$. Suppose it is true for $n = k$. Then so

$$a_k < 6$$

and

$$\frac{1}{2}(a_k + 6) < \frac{1}{2}(12) = 6$$

Thus

$$a_{k+1} < 6$$

This shows, by mathematical induction, that $a_n < 6$ for all n. Since the sequence $\{a_n\}$ is increasing and bounded, thus it has a limit. We know $L = \lim_{n\to\infty} a_n$ exists, we can use the given recurrence relation to write

$$\lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \frac{1}{2}(a_n + 6) = \frac{1}{2}\left(\lim_{n\to\infty} a_n + 6\right) = \frac{1}{2}(L + 6)$$

Since $a_n \to L$, it follows that $a_{n+1} \to L$ too (as $n \to \infty, n+1 \to \infty$ also). So we have

$$L = \frac{1}{2}(L + 6)$$

Solving this equation for L, we get $L = 6$, as we predicted.

5. (Bonus question)

(a) Prove that every convergent sequence is a Cauchy sequence.

Answer: Suppose (x_n) is a convergent sequence and it converges to L. Then, for every $\epsilon > 0$, there exists a positive integer N such that for all $n \geq N$, $|x_n - L| < \frac{\epsilon}{2}$. Now, if $m, n \geq N$, then by the triangle inequality, $|x_n - x_m| \leq |x_n - L| + |L - x_m| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$. So, (x_n) is a Cauchy sequence.

(b) The harmonic sequence is given by:

$$a_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}.$$

Is it a Cauchy sequence?

Answer: Let $\epsilon_0 = \frac{1}{2}$, for any $N > 0$, let $n \geq N$, $m = 2n$, then we have

$$|a_{2n} - a_n| = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} > \frac{1}{2n} + \frac{1}{2n} + \cdots + \frac{1}{2n} = \frac{n}{2n} = \frac{1}{2}.$$

Therefore, the harmonic sequence is not a Cauchy sequence. Because it is increasing, so it clearly diverges to infinity.